



OPERATIONAL RESEARCH & OPTIMISATION

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Outline

- Basic elements of a game
- Fundamental assumptions that underline the game theory
- Equilibria of a game
- Game theory application in public policy design

Introduction to Game Theory

- Game theory is a branch of mathematics and economics that studies strategic decision-making — how rational individuals or entities interact when their outcomes depend on each other's choices.
- John von Neumann: a founder of game theory and electronic computing; *Theory of Games and Economic Behaviour* (1944)
- Typical application: incentive mechanisms in public policy design

Game theory

- A person's decision outcome depends not only on their own choice, but also on the choices made by others they interact with.
- Elements of a game:
 - Multiple decision-makers (called players)
 - Strategies (possible actions they can take)
 - Payoffs (rewards or outcomes for each combination of actions)

Behavioural Assumptions

- Players fully understand the game structure
- Players are rational
 - Maximize their own payoff
 - Know others are rational
 - Hierarchy of beliefs

Prisoner's Dilemma

- Two suspects are interrogated separately:
- Confess vs. Deny
- Outcomes (or payoffs, the first value is the payoff for player 1, and the second value is the payoff for player 2):
 - One confesses, one denies → confessor free, other 10 years
 - Both confess → 4 years each
 - Both deny → 1 year each

Suspect 1 \ Suspect 2	Deny	Confess
Deny	-1, -1	-10, 0
Confess	0, -10	-4, -4

Best Responses Strategy

- Let **S** be a strategy chosen by Player 1, and **T** be a strategy chosen by Player 2.
A particular cell in the payoff matrix corresponds to the strategy pair (**S**, **T**).
 - $P_1(\mathbf{S}, \mathbf{T})$: the payoff that Player 1 receives from this strategy pair
 - $P_2(\mathbf{S}, \mathbf{T})$: the payoff that Player 2 receives from this strategy pair
- **Best response:**
Given Player 2's strategy **T**, if Player 1's payoff from choosing strategy **S** is greater than or equal to the payoff from choosing any other strategy, then **S** is called a *best response* to **T**.
- $P_1(S, T) \geq P_1(S', T)$,
- where **S'** denotes any other strategy of Player 1 besides **S**.

Back to Prisoner's Dilemma

- If suspect 2 choose Deny:
 - The best response strategy for Suspect 1 is Confess because $0 > -1$.
- If suspect 2 choose Confess:
 - The best response strategy for Suspect 1 is Confess because $-4 > -10$.
- Same analysis apply to Suspect 2 as well.

Suspect 1 \ Suspect 2	Deny	Confess
Deny	-1, -1	-10, 0
Confess	0, -10	-4, -4

Dominant Strategies and Strictly Dominant Strategies

- **Definitions of dominant strategies (from the perspective of best responses):**
 - A **dominant strategy S** for Player 1 is a strategy that is a best response to **every** strategy of Player 2.
 - A **strictly dominant strategy S** for Player 1 is a dominant strategy that is a **strict best response** to **every** strategy of Player 2.
- If a player has a strictly dominant strategy, we can expect that the player will choose this strategy (consistent with the basic assumptions of rationality). We call this **dominant strategy equilibrium**.
- **Note:** The concept of a dominant strategy is defined with respect to **all possible strategies of the opponent**.

Prisoner's Dilemma

Suspect 1 \ Suspect 2	Deny	Confess
Deny	-1, -1	-10, 0
Confess	0, -10	-4, -4

- The fact that it is best for Suspect 1 to choose Confess, no matter what Suspect 2 does, makes the Confess a **dominant strategy** for suspect 1.
- **Both Suspect 1 and Suspect 2 have the strictly dominant strategy of “confessing.”**
- Even though both suspects would receive lighter sentences if they both denied,
- This model illustrates that **when individual self-interest is at stake, establishing cooperation is extremely difficult.**

Exercise: Home or dancing?

- Alice and Bob are best friends in grad school. They both enjoy each other's company, but neither can communicate with the other before deciding whether to stay at home (where they would not see each other) or go swing dancing this evening (where they could see each other). Each prefers going dancing to being at home.

		Bob	
		<i>home</i>	<i>dancing</i>
Alice	<i>home</i>	(0, 0)	(0, 1)
	<i>dancing</i>	(1, 0)	(2, 2)

What do you think the players will do?

		Bob	
		<i>home</i>	<i>dancing</i>
Alice	<i>home</i>	(0, 0)	(0, 1)
	<i>dancing</i>	(1, 0)	(2, 2)

- (A) (home, home)
- (B) (home, dancing)
- (C) (dancing, home)
- (D) (dancing, dancing)

Which of the following statements is correct?

		Bob	
		<i>home</i>	<i>dancing</i>
Alice	<i>home</i>	(0, 0)	(0, 1)
	<i>dancing</i>	(1, 0)	(2, 2)

- (A) (home, home) is the only dominant strategy equilibrium.
- (B) (dancing, dancing) is the only dominant strategy equilibrium.
- (C) (dancing, home) or (home, dancing) is the only dominant strategy equilibrium.
- (D) This game has more than one dominant strategy equilibrium.
- (E) This game has no dominant strategy equilibrium.

Behavioural Reasoning in Games

- If all players have strictly dominant strategies, we can predict that they will all ***choose their strictly dominant*** strategies.
- If only one player has a strictly dominant strategy, that player will choose the strictly dominant strategy, while the other player will ***choose the best response*** to it.
- What if neither player has a strictly dominant strategy?

Nash Equilibrium



- ▶ Won Nobel prize in Economics.
- ▶ 1-page paper on Nash equilibrium and 26-page PhD thesis.
- ▶ Every finite game has at least one Nash equilibrium.

Nash Equilibrium

- Suppose Player 1 chooses strategy **S** and Player 2 chooses strategy **T**.
If **S** is a best response to **T**, and **T** is also a best response to **S**, then the strategy pair (**S**, **T**) is called a **Nash equilibrium**.
- At equilibrium, no player has an incentive (a rational reason) to change their strategy.
- A Nash equilibrium can be viewed as an **equilibrium of beliefs**:
 - The players are mutual best responses; no one can gain additional benefit by unilaterally changing their strategy, even though both players might be better off if they both changed (compared with both not changing).
 - It is **stable** and may be **mutually beneficial**.

Apply of game theory in a pricing example

- Consider two players, company A and B, that have simultaneously developed virtually identical versions of what we will call the UltraPhone. Both companies will release their UltraPhones at the same time and, therefore, each has to set a price for the phone without knowing the price set by its competitor.
- For the purposes of this example, we assume both companies will charge either a High price or a Low price for all units. Therefore, the companies' possible actions are to price 'High' or 'Low,' and each company's strategy is a plan to charge one of these two prices.

Apply of game theory in a pricing example

- To complete the description of the game, we describe the payoffs under each strategy profile.

		B Chooses...	
		High	Low
High	8	8	0
	12	0	6
Low	8	12	6
	6	6	6

Steps to identify best response, dominate strategies and Nash Equilibrium

- Given this payoff structure, what should the firms do?
- First, consider firm A's decision, given each strategy chosen by B.
 - If B chooses High (the first column), then it is better for A to choose Low. Therefore, we say that Low is player A's **best response** to B choosing High.
 - If B chooses Low (the second column), then firm A's best response is to choose Low as well.
 - The fact that it is best for A to price Low, no matter what B does, makes pricing Low a **dominant strategy**.
- Pricing Low is also a dominant strategy for B: no matter what A chooses, Low is better.
- The strategy profile [A Low, B Low] is the unique **Nash Equilibrium** of this game: if both companies choose Low, then neither has a reason to unilaterally change its mind.

A Chooses...

		B Chooses...	
		High	Low
A Chooses...	High	8 8 8 0	12 0 0 6
	Low	12 0 12 6	6 6 6 6

Observations

- Because we assume that both companies are rational and fully anticipate the logic used by the other player (the infinite belief hierarchy), we predict that both will charge Low, each will earn 6, and the total industry contribution margin will be 12.
- This margin is less than the optimal result of 16. This is a well-known result: under competition, both prices and industry profits decline.
- However, that it may be possible to achieve higher profits when the game is played repeatedly.

Nash Equilibrium

- The strategy profile [A Low, B Low] is a Nash equilibrium in **pure strategies**.
- It is a pure strategy equilibrium because each player chooses a single price. The players might also consider a **mixed strategy**, in which each player chooses a probability distribution over a set of pure strategies,
- The **equilibrium concept** is named after **John Nash**, a mathematician who showed that if each player has a finite number of pure strategies, then there must always exist at least one mixed strategy or pure strategy equilibrium (Nash, 1950)

Nash Equilibrium

- In certain games, however, there may not exist a pure strategy equilibrium.
- It is also possible to have multiple pure-strategy equilibria.

Multiple Equilibria and pricing example with customers loyalty

- Now return to the UltraPhone pricing game with two price points(High, Low), although we also incorporate product differentiation, but now we assume customers have loyalty.
- Assume that 40 percent of the High-segment customers are loyal to A and will choose A's product even if A chooses a High price while B chooses a Low price.
- Likewise, 40 percent of the High-segment customers are loyal to B. The payoffs from this new game are shown in the Table.

Multiple Equilibria and pricing example with customers loyalty

- The presence of loyal customers significantly changes the behaviour of the players.
- If B chooses High, then A's best response is to choose Low, while if B chooses Low, A's best response is to choose High.
- In this game, neither company has a dominant strategy, and [A Low, B Low] is no longer an equilibrium; if the companies find themselves in that cell, companies would have an incentive to move to High. But [A High, B High] is not an equilibrium either; both would then want to move to Low.

		B Chooses...	
		High	Low
A Chooses...	High	8 8	9.6 6.4
	Low	6.4 9.6	6 6

Multiple Equilibria and pricing example with customers loyalty

- In fact, this game has **two pure strategy Nash equilibria**: [A High, B Low] and [A Low, B High].
- Given either combination, neither company has any incentive to unilaterally move to another strategy. Each of these two equilibria represents a split in the market between one company that focuses on its own High-end loyal customers and the other that captures everyone else.

		B Chooses...	
		High	Low
A Chooses...	High	8 8	9.6 6.4
	Low	6.4 9.6	6 6

Multiple Equilibria

- The existence of multiple equilibria poses a challenge for both theorists and practitioners who use game theory.
- Given the many possible equilibria, will players actually choose one?
- If so, which one?
- Behavioural explanations have been proposed to answer this question. In addition, game theorists have invented variations of the Nash equilibrium to address these questions. These variations are called **refinements**, and they lay out criteria that exclude certain types of strategies and equilibria.
- Many of these innovations depend upon the theory of **dynamic games**, in which the game is played in multiple stages over time (Govindan and Wilson, 2005).

Exercise: Dancing or running?

		Bob	
		<i>dancing</i>	<i>running</i>
Alice	<i>dancing</i>	(2, 2)	(0, 0)
	<i>running</i>	(0, 0)	(1, 1)

Alice and Bob would like to sign up for an activity together.

They both prefer dancing over running.

They also prefer signing up for the same activity over signing up for two different activities.

Which of the following statements is correct?

		Bob	
		<i>dancing</i>	<i>running</i>
Alice	<i>dancing</i>	(2, 2)	(0, 0)
	<i>running</i>	(0, 0)	(1, 1)

- (A) (dancing, dancing) is the only dominant strategy equilibrium.
- (B) (running, running) is the only dominant strategy equilibrium.
- (C) (dancing, running) or (running, dancing) is the only dominant strategy equilibrium.
- (D) This game has more than one dominant strategy equilibrium.
- (E) This game has no dominant strategy equilibrium.

Which of the following is correct?

Consider only pure-strategy Nash equilibria.

		Bob	
		<i>dancing</i>	<i>running</i>
Alice	<i>dancing</i>	(2, 2)	(0, 0)
	<i>running</i>	(0, 0)	(1, 1)

- (A) (dancing, dancing) is the only Nash equilibrium.
- (B) (running, running) is the only Nash equilibrium.
- (C) (dancing, dancing) and (running, running) are both Nash equilibria.
- (D) This game has more than two Nash equilibria.

Pareto dominance and Pareto optimality

- **Pareto dominance:**
- An outcome o Pareto dominates another outcome o' if and only if
 - every player is weakly better off in o and
 - at least one player is strictly better off in o .
- **A Pareto optimal outcome:**
- An outcome o is Pareto optimal if and only if no other outcome o' Pareto dominates o .
- **What is the difference between the statements below?(which one is the definition of Pareto optimal?)**
 - An outcome o Pareto dominates all other outcomes.
 - An outcome o is NOT Pareto dominated by any other outcome.

Exercise: Dancing or running? Which outcome is Pareto Optimal?

		Bob	
		<i>dancing</i>	<i>running</i>
Alice	<i>dancing</i>	(2, 2)	(0, 0)
	<i>running</i>	(0, 0)	(1, 1)

- (A) (dancing, dancing)
- (B) (dancing, running)
- (C) (running, dancing)
- (D) (running, running)

Go back to Prisoner's Dilemma

Suspect 1 \ Suspect 2	Deny	Confess
Deny	-1, -1	-10, 0
Confess	0, -10	-4, -4

- **Both Suspect 1 and Suspect 2 have the strictly dominant strategy of “confessing.”**
- **(Confess, confess) is also Nash equilibrium.**
- **This (Confess, confess) is not Pareto optimal.** i.e. both suspects would receive lighter sentences if they both denied.
- This model illustrates that **when individual self-interest is at stake, establishing cooperation is extremely difficult.**

Public policy design

- Public policy design can be effectively explained through the lens of game theory, which studies strategic decision-making in situations where the outcome for each participant depends upon the choices made by others.
- **Pareto optimality:** Governments and planners aim for social efficiency.
- **Nash equilibrium** describes how people behave. A Nash equilibrium can be inefficient and socially undesirable.

Mechanism Design

- *“Policies can shape incentives and interactions to guide individual self-interested choices toward a desirable collective outcome.”*
- Policy is a tool for *mechanism design*, used to change the structure of the game and create incentives for cooperation and socially beneficial behaviour.
- Policy designers modify the rules to achieve a socially optimal equilibrium.
- **Intuition:**
 - **Pareto optimality** → *What outcome do we want?*
 - **Nash equilibrium** → *What outcome will occur given incentives?*
 - **Policy design** → *Change incentives so the desired outcome is a Nash equilibrium*

A simple tax policy example: income tax and tax evasion

- **Setup**

- Citizens choose whether to **report income honestly** or **evade taxes**
- Government chooses **tax rate and enforcement**
- Public goods (roads, healthcare) are funded by taxes

A simple tax policy example: income tax and tax evasion

- **Nash Equilibrium (What Happens Without Policy Design)**
- Suppose:
 - Tax rates are high
 - Enforcement is weak
- **Individuals' incentives:**
 - If others pay taxes, I gain by evading
 - If others evade, I also gain by evading
- **Nash equilibrium:** widespread **tax evasion**
- **Result:**
 - Low tax revenue
 - Poor public services
 - Everyone worse off than if most people complied
- This Nash equilibrium is **Pareto inefficient**
- (add a value example)

A simple tax policy example: income tax and tax evasion

- **Pareto-Optimal Outcome (What Society Wants)**
- A **Pareto-optimal** tax outcome would have:
 - High compliance
 - Adequate revenue
 - Well-funded public goods
- In this outcome:
 - Citizens get better services
 - No one is worse off compared to the evasion equilibrium (better infrastructure, stability, social insurance)
- This outcome is **not automatic** — it doesn't arise on its own.

Role of Tax Policy: Aligning Incentive

- The government **redesigns the game** by introducing:
 - Moderate tax rates
 - Strong enforcement (audits, penalties)
 - Simplified tax filing
 - Social norms and transparency
- **Effect on Incentives**
- Now:
 - Evading taxes becomes risky and costly
 - Paying taxes becomes the **best response strategy**
- **New Nash equilibrium:** tax compliance
- This Nash equilibrium is also **Pareto optimal**
- (add a value example)

Other examples of policy interventions

Policy Tool 	Game Theory Principle	Mechanism of Action	Example
Taxes & Subsidies	Altering Payoffs	Change the costs/benefits associated with a specific action to favor socially optimal strategies [4].	Carbon taxes make polluting more expensive, encouraging firms to reduce emissions.
Regulation & Laws	Constraining Strategies	Limit the available choices or impose penalties for undesirable actions [4].	Bans on certain types of fishing prevent the tragedy of the commons in shared water bodies.
Providing Information	Changing Perceptions	Ensure players understand the true consequences of their actions on themselves and others, potentially fostering cooperation [4].	Public health campaigns about the benefits of vaccination.
Establishing Property Rights	Redefining Resources	Clearly defining ownership of resources prevents overuse by making individuals responsible for the asset's upkeep [3].	Creating individual fishing quotas to manage shared fish stocks.

- *Science policies as principal–agent games: Institutionalization and path dependency in the relation between government and science*, Barend Van der Meulen, Research Policy, 1998, 397-414

- Q&A